Resolution Geometry on a Real Seismic Catalogue

A Necessity Test with Honest Nulls

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Abstract

The necessity argument for resolution geometry (RG) was, so far, demonstrated on synthetic data. This note takes it to a real, neutral, public dataset—a catalogue of 782 significant global earthquakes—and adds the ingredient the earlier CTMT seismic run lacked: proper permutation and bootstrap nulls, so that a claimed effect can be separated from sampling noise. The result is deliberately mixed and reported without spin. **Pillar 1 (coupling)** holds and is cross-validated: treating the physical source features and the felt-impact features as one covariance object recovers 40% of the impact variance from the source, and predicts held-out impact with out-of-sample R^2 up to 0.54; independent per-feature probes cannot do this. We concede this pillar's classical root (conditional Gaussian), so it confirms the coupling is real and useful without being RG-exclusive. **Pillar 2 (holonomy)**—the RG-specific one—is an *honest negative* on this catalogue: the resolved feature-frame transported around the longitude cycle produces a holonomy that does *not* exceed a permutation null ($p = 0.20$–0.66). The diagnosis is clean: disjoint longitude bins of a few hundred large events vary sharply, not smoothly, between neighbours (real adjacent-frame overlap 0.44, below the shuffled 0.74), and a closed-loop holonomy needs smooth variation. That the null test refuses to certify a holonomy here is itself the point: RG does not manufacture curvature. Crucially, the required smoothness *does* exist along physical axes (overlapping-window adjacent overlap 0.93–0.98 in depth, magnitude, latitude), so the obstruction is data volume, not absence of structure. **Pillar 3 (invariants)** is supported: the coupling invariants are comparable across regions where raw loadings are not. The honest conclusion: on this catalogue the demonstrable necessity is the coupling; a genuine RG-specific holonomy test requires a magnitude-complete regional catalogue binned densely enough to form a smooth closed loop—the sharpened form of the "next experiment."

Contents

1 Purpose and relation to the CTMT seismic note

An earlier note tested the CTMT morphism family on a global earthquake catalogue and a six-event local declustered chart, reporting monodromy "defect rates" across annual regimes. Two things were missing. First, a *null baseline*: without one, a defect rate cannot be told from what sampling noise alone would produce in small annual charts. Second, the clean RG vocabulary: resolved/null sectors, canonical correlations, and the transport holonomy, rather than an ad hoc diagnostic list. This note supplies both, on the same public data, and asks the sharper question the necessity argument requires: *do the RG necessity phenomena appear in real seismic data above noise?* The answer is reported without spin, positive where the data support it and negative where they do not. No claim about earthquake physics or prediction is made or implied; the catalogue is a neutral test object.

Data. 782 significant events ($M \geq 6.5$, 2001–2022) with magnitude, a log-energy proxy, depth, felt intensity (CDI, MMI), significance, tsunami flag, azimuthal gap, station count, and minimum distance. Features are robustly standardized (median/MAD, clipped at 6). Geographic coordinates are used as *conditions*, not features.

2 Pillar 1 — Coupling: the impact sector is recoverable from the source sector

Split the features into a *source* sector (magnitude, log-energy, depth, gap, station count, distance—what instruments record) and an *impact* sector (CDI, MMI, significance, tsunami—the felt outcome one would like to infer). Treating them as one covariance object, the coupling is strong and graded:

$$\text{canonical correlations (source} \leftrightarrow \text{impact)} = (0.72,\ 0.61,\ 0.40,\ 0.15).$$

Finding 1 (Coupling recovers the un-probed sector, out of sample)**.** The Schur complement gives an impact-sector uncertainty of 0.60 against a prior of 1.00: 40% of the felt-impact variance is recoverable from source features via the coupling. On held-out events (train 600, test 182) the recovery is genuine, not overfit:

impact feature	CDI	MMI	significance	tsunami
out-of-sample R^2	0.19	0.32	0.44	0.54

Independent per-feature probes, having no cross-sector model, cannot recover any of this.

Remark 1 (Conceded root). This pillar's core is conditional-Gaussian inference; it is not RG-exclusive. It confirms on real data that the coupling is real and useful—one facet of the coherent object—but it does not, by itself, establish the RG-specific necessity. That burden falls on Pillar 2.

3 Pillar 2 — Holonomy: an honest negative, correctly reported

The RG-specific necessity is the transport holonomy: if the observation geometry varies across a cycle of conditions with curvature, naive cross-condition comparison carries a bias invisible to local checks. Longitude is a naturally cyclic condition (the Ring of Fire closes on itself), so we partition events into K longitude sectors, build the resolved feature-frame per sector, and transport it around the cycle.

Finding 2 (No holonomy above the null on this catalogue)**.** The observed holonomy does not exceed a permutation null (labels shuffled across sectors, 2000 draws):

K	events/sector	observed holonomy	null 95th pct	p-value	real vs shuffled adj. overlap
6	130	1.17	3.14	0.66	0.44 vs 0.74
8	97	3.14	3.14	0.20	0.48 vs 0.73
12	65	3.14	3.14	0.40	0.53 vs 0.61

The diagnosis is in the last column. Shuffling makes each sector a random draw from the whole catalogue, so all sectors inherit the *same* global covariance and their frames align (overlap 0.74). The real sectors align *less* (0.44)—genuine regional structure—but that structure changes *sharply* between neighbouring longitude bins, not smoothly. A closed-loop holonomy is only well defined when the frame varies smoothly; here it does not, and the transport is dominated by large jumps, saturating near π and indistinguishable from noise.

Finding 3 (The required smoothness exists—along physical axes)**.** Ordering events by a physical axis and using overlapping windows, the resolved frame *does* vary smoothly:

axis	depth	magnitude	latitude
mean adjacent overlap	0.93	0.96	0.98

So the smooth cross-condition structure Pillar 2 needs is present in the data; what the 782-event catalogue lacks is enough events to fill a *cyclic* partition (or a 2D condition grid) densely enough to trace a smooth *closed* loop.

Remark 2 (The integrity of the negative). This is the self-limiting property working on real data. With a proper null, RG does not report a holonomy that the data cannot support—exactly the behaviour that separates a diagnostic from a device that manufactures structure. The earlier "defect rate" numbers, read without a null, risked the opposite. The honest state is: no RG-specific holonomy is demonstrable on this catalogue, and RG says so.

4 Pillar 3 — Invariants: comparable across regions where raw loadings are not

Splitting the catalogue into two broad regions (Americas, lon < -30; Asia/Oceania, lon > 90), the raw leading loadings differ while the coupling invariants remain comparable:

region	raw leading loading$_1$	source–impact canonical correlations
Americas	-0.052	$(0.69, 0.64, 0.39, 0.16)$
Asia/Oceania	-0.043	$(0.76, 0.67, 0.44, 0.16)$

The invariant summary is the one that travels between regions; raw loadings do not. This supports Pillar 3 on real data, modestly (two regions, not a formal multi-site study).

5 What a genuine Pillar-2 test requires

The obstruction here is volume and partition, not absence of structure (Findings 2–3). A real RG-specific holonomy test needs:

- a magnitude-complete *regional* catalogue (down to the completeness magnitude), so per-bin counts reach several hundred to stabilize the frame;

- a *cyclic* condition (e.g. position along a subduction arc, or a longitude ring in a dense region) binned densely enough that neighbouring frames overlap smoothly, or a 2D physical condition grid (e.g. depth $\times$ along-arc position) traversed as a rectangle;
- the permutation and bootstrap nulls used here, as the acceptance test.

This is the sharpened version of the earlier note's "next experiment," with the null baseline made mandatory.

6 Honest verdict

pillar	on this real catalogue	status
1 coupling (information)	OOS R^2 up to 0.54; 40% recovered	positive (classical root)
2 holonomy (consistency)	not above permutation null	honest negative; needs denser data
3 invariants (reproducibility)	comparable across regions	supported (modest)

On this catalogue the demonstrable necessity is the coupling, which is real and cross-validated but classically rooted. The RG-specific holonomy is *not* yet demonstrable here—and the honest, null-tested reporting of that absence, together with the evidence that the required smooth structure exists along physical axes, is the credible foundation for the larger test. RG earns trust here not by finding an effect but by declining to invent one.

Non-claims

Non-claim 1. No earthquake prediction, no tectonic law, no seismic-physics claim. The catalogue is a neutral test object for a methodological necessity argument.

Non-claim 2. The coupling result is conditional-Gaussian inference made one facet of the RG object; it does not claim novelty. The six-event local declustered chart is illustrative, not a statistical basis for any claim.

Non-claim 3. The Pillar-2 negative is specific to this catalogue and partition; it is not evidence that seismic feature geometry is globally flat, only that this dataset cannot resolve a closed loop.

References

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